

CENGİZ TOPEL, Turkey

WMO: 172005

Lat: 40.735N

Lon: 30.083E

Elev: 56

StdP: 100.66

Time Zone: 3.00 (E03)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.0	-3.1	-6.0	2.3	-2.2	-4.2	2.7	-0.1	8.8	14.9	7.8	12.7	0.7	320	0.423

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.7	33.2	22.3	31.9	22.1	30.8	21.9	24.7	29.4	24.0	28.7	23.2	27.9	3.8	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.2	18.1	26.6	22.8	17.6	26.4	21.9	16.7	25.8	75.1	29.4	72.3	28.9	69.2	27.5	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.6	6.4	5.5	-6.7	36.8	2.3	2.4	-8.3	38.5	-9.7	39.8	-11.0	41.2	-12.6	42.9		
			-7.4	26.0	2.4	1.3	-9.1	26.9	-10.4	27.6	-11.8	28.3	-13.5	29.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.5	5.6	6.6	8.9	12.3	17.1	21.1	23.3	24.1	20.1	15.9	11.4	7.3
	DBStd	7.30	4.37	4.46	4.00	3.52	3.07	2.59	2.13	2.31	2.72	3.56	3.80	4.35
	HDD10.0	483	148	114	68	14	1	0	0	0	0	3	27	107
	HDD18.3	1922	393	330	293	183	61	4	0	0	13	92	210	342
	CDD10.0	2137	13	18	35	84	221	333	413	436	304	184	69	25
	CDD18.3	534	0	1	1	3	23	88	156	178	68	15	2	1
	CDH23.3	5427	1	2	12	71	299	916	1601	1749	616	138	21	3
	CDH26.7	1918	0	0	1	13	73	308	602	706	181	32	2	0
Wind	WSAvg	2.0	1.9	2.2	2.3	2.2	2.1	2.1	2.3	2.2	2.0	1.7	1.6	1.7
Precipitation	PrecAvg	798	90	76	73	59	50	63	36	41	54	81	73	104
	PrecMax	1093	167	168	119	143	145	208	115	200	137	195	144	239
	PrecMin	550	23	17	31	4	6	2	1	0	0	9	14	43
	PrecStd	139	39	39	24	30	34	47	32	43	38	47	38	49
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.2	22.2	25.1	28.2	31.1	34.2	35.1	35.8	33.2	29.8	26.0	22.9
		MCWB	11.9	13.2	13.7	17.0	20.2	22.6	22.9	22.4	20.7	18.4	16.5	13.4
	2%	DB	17.8	19.1	22.0	25.2	28.8	31.2	33.0	33.6	30.9	26.9	22.9	19.4
		MCWB	11.2	11.4	12.6	15.9	19.6	21.4	22.7	22.4	20.7	18.5	15.5	12.4
	5%	DB	15.4	17.0	19.8	22.9	26.8	30.0	31.2	31.9	28.8	24.2	20.2	17.1
		MCWB	10.0	10.8	12.1	14.9	18.7	21.3	22.1	22.8	20.3	17.8	14.6	11.6
	10%	DB	13.0	14.2	17.0	20.1	24.8	28.2	30.0	30.2	26.9	22.1	18.0	14.8
		MCWB	9.1	9.8	11.1	14.0	18.0	20.8	21.8	22.4	20.0	17.3	13.8	10.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.1	14.3	14.9	18.3	22.2	24.6	25.6	25.7	24.0	21.0	18.5	14.7
		MCDB	17.4	20.4	22.2	25.5	27.4	30.5	30.8	30.5	27.9	24.2	22.7	19.7
	2%	WB	11.6	12.4	13.5	16.7	20.8	23.5	24.4	24.9	22.6	20.1	17.0	13.3
		MCDB	16.5	17.5	19.7	23.1	26.1	28.6	29.3	29.6	26.9	23.2	20.2	17.8
	5%	WB	10.4	11.2	12.6	15.5	19.7	22.5	23.6	24.2	21.5	18.9	15.5	12.2
		MCDB	14.2	15.9	18.3	21.4	24.6	27.6	28.6	28.8	26.2	23.0	18.9	16.1
	10%	WB	9.4	10.2	11.5	14.3	18.7	21.6	22.9	23.5	20.7	18.0	14.2	11.0
		MCDB	12.7	14.3	16.5	19.5	23.3	26.4	27.6	28.1	25.1	21.5	17.5	14.0
Mean Daily Temperature Range		MDBR	8.9	9.3	11.1	12.4	12.9	13.0	13.1	12.7	12.7	11.2	10.6	9.7
	5% DB	MCDBR	13.0	14.2	17.1	18.1	17.0	15.9	15.4	15.0	16.8	15.2	14.6	13.2
		MCWBR	7.3	7.8	8.8	9.6	8.7	7.5	6.8	6.0	7.8	7.8	8.2	7.3
	5% WB	MCDBR	11.6	12.9	14.2	15.6	14.3	13.7	13.0	12.5	13.0	11.6	12.2	11.2
		MCWBR	6.7	7.4	7.8	8.7	8.0	7.1	6.4	5.6	6.6	6.3	7.5	6.5
Clear-Sky Solar Irradiance	taub	0.359	0.388	0.430	0.471	0.441	0.426	0.430	0.436	0.423	0.403	0.369	0.359	
	taud	2.326	2.229	2.127	2.036	2.196	2.284	2.289	2.269	2.291	2.322	2.362	2.348	
	Ebn at Noon	778	807	812	805	841	852	845	828	813	784	762	745	
	Edh at Noon	90	114	140	164	143	132	130	129	118	103	86	82	
All-Sky Solar Radiation	RadAvg	1.55	2.30	3.49	4.77	5.98	6.84	7.08	6.20	4.64	3.08	2.08	1.45	
	RadStd	0.22	0.33	0.33	0.54	0.54	0.60	0.27	0.41	0.37	0.27	0.27	0.21	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (12 neighbors)	+0.48	N/A	N/A	N/A	N/A	N/A	N/A	-90	N/A	N/A

Nomenclature: See separate page